SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Code of Conduct

I Tamil Elakiya S (192524443) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tamil Elakiya S

Digital farmer Marketplace for Agricultural Product Trading

1. Identify the Project Vision and Stakeholders

1.1 Project Vision

The Digital Farmer Marketplace is a web-based platform designed to modernize agricultural product trading by establishing a direct connection between farmers and buyers. The vision of the project is to create a transparent, secure, and efficient marketplace that eliminates unnecessary intermediaries, promotes fair pricing, and improves the overall agricultural supply chain.

In many traditional agricultural markets, farmers depend on middlemen to sell their produce. This often leads to lower profits for farmers while buyers pay higher prices. The proposed platform addresses this issue by providing a centralized digital marketplace where farmers can directly list their products and buyers can purchase them through an easy-to-use online system.

The platform also aims to encourage digital transformation in agriculture by simplifying product management, order processing, payment handling, and communication between stakeholders. By adopting Agile software development practices, the project can continuously improve based on stakeholder feedback and changing market requirements.

1.2 Project Objectives

The major objectives of the proposed system are:

- Develop a secure online marketplace for agricultural products.
- Connect farmers directly with consumers and other buyers.
- Eliminate unnecessary intermediaries involved in agricultural trading.
- Provide transparent and competitive product pricing.
- Enable secure digital payment processing.
- Simplify product listing, order placement, and inventory management.

- Reduce food wastage through quicker product sales.
- Maintain digital records of users, products, transactions, and orders.
- Improve accessibility for farmers in different regions.
- Build a scalable and maintainable software system using Agile principles.

1.3 Key Stakeholders

The successful implementation of the Digital Farmer Marketplace depends on the active participation of several stakeholders. Each stakeholder contributes to different stages of the trading process and benefits from the system in different ways.

1. Farmers

Farmers are the primary users of the system. They register on the platform, upload agricultural products, manage inventory, receive customer orders, and obtain payments. The system enables them to reach a larger customer base while receiving fair prices for their produce.

2. Buyers / Consumers

Buyers include individual consumers, households, and institutions that purchase agricultural products. They can search products, compare prices, place orders, complete payments, and provide ratings and feedback after receiving products.

3. Retailers and Wholesalers

Retailers and wholesalers purchase agricultural products in bulk for resale. The marketplace helps them identify reliable farmers, compare product quality, and establish long-term business relationships.

4. Hotels and Restaurants

Hotels, restaurants, and catering services require fresh agricultural products regularly. The platform enables them to source quality products directly from farmers at competitive prices.

5. Delivery and Logistics Partners

Delivery partners are responsible for transporting products safely from farmers to buyers. They update delivery status and ensure that products reach customers on time, particularly for perishable goods.

6. Platform Administrator

The administrator manages the overall functioning of the marketplace. Responsibilities include verifying users, monitoring products, handling complaints, generating reports, and maintaining system security.

7. Banks and Payment Gateway Providers

Banks and payment gateways process online transactions securely. They ensure safe transfer of payments between buyers and farmers while maintaining financial security.

8. Government and Agricultural Departments

Government agencies support digital agriculture initiatives, promote fair trade practices, and encourage farmers to adopt digital platforms. They may also use the system to monitor agricultural market trends and implement farmer welfare programs.

1.4 Stakeholders Responsibilities

Stakeholder	**Responsibilities**
Farmers	Register, upload products, manage inventory, fulfill orders
Buyers	Search products, place orders, complete payments, provide feedback
Retailers & Wholesalers	Purchase products in bulk and maintain supply chains
Hotels & Restaurants	Procure fresh agricultural products regularly
Logistics Partners	Deliver products safely and update delivery status
Administrator	Manage users, products, transactions, and reports
Banks/Payment Gateways	Process secure online payments
Government Agencies	Promote digital agriculture and monitor fair trading

1.5 Stakeholder Interaction

The Digital Farmer Marketplace acts as a centralized platform connecting all stakeholders. Farmers upload products to the marketplace, buyers browse and purchase them, payment gateways process transactions, logistics partners handle product delivery, and administrators monitor the complete process. Government agencies can use analytical reports generated by the platform to understand agricultural market trends and support policy-making.

This collaborative interaction improves transparency, reduces communication gaps, and creates a more efficient agricultural trading ecosystem.

1.6 Expected Business Value

- The proposed project creates value for every stakeholder involved in the agricultural supply chain.
- Farmers receive better prices and increased market access.
- Buyers obtain fresh products at fair prices.
- Retailers and wholesalers experience simplified procurement.
- Delivery partners gain additional business opportunities.
- Administrators efficiently manage marketplace operations.
- Government agencies obtain better market data for agricultural planning.

2. Create Epics and User Stories

2.1 Introduction to Epics and User Stories

In Agile software development, requirements are divided into **Epics** and **User Stories** to simplify planning and implementation. An **Epic** represents a large feature or business functionality, while a **User Story** describes a specific requirement from the perspective of an end user. Breaking the project into smaller user stories helps the development team prioritize tasks, monitor progress, and deliver functional software incrementally.

For the Digital Farmer Marketplace, the requirements are categorized into several epics covering user management, product management, marketplace operations, payment processing, delivery management, and system administration.

Epic 1: User Management

This epic focuses on managing user accounts, authentication, and profile management for farmers, buyers, and administrators.

User Story 1

As a farmer, I want to register on the platform so that I can sell my agricultural products online.

User Story 2

As a buyer, I want to create an account so that I can purchase agricultural products securely.

User Story 3

As a registered user, I want to log in securely so that I can access my account safely.

User Story 4

As a user, I want to update my profile information so that my account details remain accurate.

Epic 2: Product Management

This epic manages agricultural product information uploaded by farmers.

User Story 5

As a farmer, I want to add new agricultural products so that buyers can view and purchase them.

User Story 6

As a farmer, I want to upload product images so that buyers can evaluate product quality.

User Story 7

As a farmer, I want to update product quantity and price so that my listings remain current**.**

User Story 8

As a farmer, I want to remove unavailable products so that buyers only see available items.

Epic 3: Marketplace and Order Management

This epic supports product browsing, purchasing, and order management.

User Story 9

As a buyer, I want to search products by name or category so that I can quickly find what I need.

User Story 10

As a buyer, I want to filter products based on price and location so that I can make better purchasing decisions.

User Story 11

As a buyer, I want to place an order online so that I can purchase products directly from farmers.

User Story 12

As a buyer, I want to track my order status so that I know when my products will be delivered.

Epic 4: Payment Management

This epic manages secure financial transactions between buyers and farmers.

User Story 13

As a buyer, I want to make secure online payments so that my transactions are completed safely.

User Story 14

As a farmer, I want to receive payment confirmation so that I know when payments have been received.

User Story 15

As a user, I want to view my transaction history so that I can keep track of previous payments.

Epic 5: Delivery and Logistics Management

This epic focuses on delivering agricultural products efficiently.

User Story 16

As a delivery partner, I want to receive delivery requests so that I can transport products to buyers**.**

User Story 17

As a buyer, I want to receive delivery updates so that I can know the status of my order.

User Story 18

As a farmer, I want to know when my products have been delivered so that I can complete the order successfully.

Epic 6: Administration and Reporting

This epic allows administrators to manage the platform and monitor its activities.

User Story 19

As an administrator, I want to verify user accounts so that only genuine users access the platform.

User Story 20

As an administrator, I want to monitor transactions so that fraudulent activities can be prevented.

User Story 21

As an administrator, I want to manage product listings so that inappropriate or incorrect information can be removed.

3. Define Acceptance Criteria

3.1 Introduction

Acceptance criteria define the conditions that must be satisfied before a user story is considered complete. They help developers, testers, and stakeholders clearly understand the expected behavior of the system. In Agile development, acceptance criteria are commonly written using the Given–When–Then format, which improves communication and simplifies software testing.

For the Digital Farmer Marketplace, the following acceptance criteria are defined for the most important user stories.

User Story 1: Farmer Registration

User Story

As a farmer, I want to register on the platform so that I can sell my agricultural products online.

Acceptance Criteria

- Given the farmer is on the registration page
- When valid personal and farming details are entered
- Then the farmer account should be successfully created and redirected to the login page.

User Story 2: Buyer Registration

User Story

As a buyer, I want to create an account so that I can purchase agricultural products online.

Acceptance Criteria

- Given the buyer opens the registration page
- When valid information is submitted
- Then the buyer account should be created successfully.

User Story 3: User Login

User Story

As a registered user, I want to log into the system so that I can access my account securely.

Acceptance Criteria

- Given the user has a registered account
- When the correct username and password are entered
- Then the user should successfully access the dashboard.

User Story 4: Product Listing

User Story

As a farmer, I want to upload my agricultural products so that buyers can purchase them.

Acceptance Criteria

- Given the farmer is logged in
- When product details, price, quantity, and image are submitted
- Then the product should be displayed in the marketplace.

User Story 5: Product Search

User Story

As a buyer, I want to search agricultural products so that I can easily find the required items.

Acceptance Criteria

- Given products are available in the marketplace
- When the buyer searches using a product name or category
- Then matching products should be displayed.

User Story 6: Place Order

User Story

As a buyer, I want to place an order so that I can purchase products directly from farmers.

Acceptance Criteria

- Given the buyer selects an available product
- When the order is confirmed
- Then the order should be created successfully and visible in both buyer and farmer dashboards.

User Story 7: Online Payment

User Story

As a buyer, I want to pay online so that I can complete my purchase securely.

Acceptance Criteria

- Given the order has been placed
- When the payment is completed successfully
- Then the payment should be recorded and a confirmation message should be displayed.

User Story 8: Order Tracking

User Story

As a buyer, I want to track my order so that I know its delivery status.

Acceptance Criteria

- Given the order has been dispatched
- When the buyer opens the order details
- Then the current delivery status should be displayed.

User Story 9: User Verification

User Story

As an administrator, I want to verify registered users so that only genuine users access the system.

Acceptance Criteria

- Given a new registration request exists
- When the administrator approves the account
- Then the user should receive permission to use the platform.

User Story 10: Report Generation

User Story

As an administrator, I want to generate reports so that I can monitor system performance and transactions.

Acceptance Criteria

- Given transaction records are available
- When the administrator selects a report type
- Then the system should generate an accurate report.

4. Prioritize the Product Backlog

4.1 Introduction

The Product Backlog is a prioritized list of all the features, enhancements, bug fixes, and requirements that need to be developed for the Digital Farmer Marketplace. In Agile Scrum, the Product Owner is responsible for maintaining and prioritizing the backlog based on business value, customer needs, implementation dependencies, and project goals.

For the proposed system, the product backlog is prioritized so that the most essential features required for the basic functioning of the marketplace are developed first. Features that directly benefit farmers and buyers are given the highest priority, while additional features that improve user experience are scheduled for later sprints.

4.2 Product Backlog

Product Backlog Item	Priority	Business Value	Dependency
Farmer Registration	High	Very High	None
Buyer Registration	High	Very High	None
User Login & Authentication	High	Very High	Registration
Product Listing	High	Very High	Farmer Login
Product Search & Filter	High	Very High	Product Listing
Order Placement	High	Very High	Product Search
Payment Integration	High	High	Order Placement
Order Tracking	Medium	High	Order Placement
User Profile Management	Medium	Medium	Login
Delivery Management	Medium	Medium	Order Placement
Notifications	Medium	Medium	Orders & Payments
Product Reviews & Ratings	Low	Medium	Completed Orders

Product Backlog Item	Priority	Business Value	Dependency
Sales Reports	Low	Medium	Transactions
Analytics Dashboard	Low	Medium	Sales Reports

4.3 Priority Classification

High Priority (Must Have)

These features are essential for the first working version of the application. Without them, the marketplace cannot function effectively.

The high-priority features include:

- Farmer Registration
- Buyer Registration
- User Login
- Product Listing
- Product Search
- Order Placement
- Payment Integration

Medium Priority (Should Have)

Medium-priority features improve system efficiency and user experience but are not mandatory for the initial release.

These include:

- Order Tracking
- Delivery Management
- User Profile Management
- Notifications

Low Priority (Could Have)

These features enhance the overall usability of the platform but are not essential during the early development phases.

Examples include:

- Product Ratings and Reviews
- Sales Reports
- Analytics Dashboard

These features can be introduced in future sprints after the primary marketplace functions are completed.

4.4 Prioritization Technique

The MoSCoW Prioritization Technique is used to organize the product backlog.

Must Have

Core functionalities that are necessary for the system to operate.

- User Registration
- Login
- Product Listing
- Product Search
- Order Placement
- Payment System

Should Have

Important features that improve usability and customer satisfaction.

- Order Tracking
- Notifications
- Delivery Management
- User Profile Management

Could Have

Additional features that enhance user experience if time permits.

- Product Reviews
- Sales Reports
- Analytics Dashboard

Won't Have (Current Sprint)

These features are planned for future releases and are excluded from the current sprint.

- AI-based Crop Price Prediction
- Weather Forecast Integration
- Multilingual Support
- Blockchain-based Transactions
- Mobile Application
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4.5 Justification for Product Backlog Prioritization

The product backlog is prioritized according to business value, user requirements, and feature dependencies. Registration and authentication are developed first because every other feature depends on authenticated users. Product listing and searching are prioritized next, as they enable farmers to sell products and buyers to browse available items. Order placement and payment integration follow because they complete the buying and selling process.

Additional features such as notifications, delivery tracking, analytics, and product reviews are scheduled for later sprints since they enhance the system but are not essential for the initial working version. This prioritization strategy ensures that the development team delivers a functional product early while allowing future improvements based on user feedback.

5. Plan the Sprint Backlog

5.1 Introduction

A Sprint Backlog is a list of user stories and development tasks selected from the product backlog for completion during a sprint. It helps the Scrum team focus on delivering a working product increment within a fixed time period. For the Digital Farmer Marketplace, the first sprint focuses on developing the core functionalities required to establish the basic marketplace.

Sprint Details

- **Sprint Duration:** 2 Weeks
- **Sprint Goal:** Develop the core modules of the Digital Farmer Marketplace.
- **Sprint Team:** Product Owner, Scrum Master, Developers, Testers, UI/UX Designer.

Sprint Backlog

User Story	Development Tasks
Farmer Registration	Design registration page, validate user inputs, store farmer details in the database
Buyer Registration	Develop signup page and buyer profile management
User Login	Implement secure authentication and password validation
Product Listing	Develop product upload form and inventory management
Product Search	Implement search and filter functionality
Admin Dashboard	Create dashboard to manage users and products

Sprint Activities

- Design the user interface for registration and login.
- Develop the database for storing users and product details.
- Implement authentication and authorization.
- Create product listing and search modules.
- Perform unit testing and fix identified issues.
- Review completed tasks at the end of the sprint.

Expected Outcome

At the end of Sprint 1, the system will provide a functional marketplace where users can register, log in, upload products, search products, and administrators can manage users through a basic dashboard. This serves as the foundation for future sprints.

6. Estimate Effort Using Story Points

6.1 Introduction

Story points are used in Agile to estimate the effort required to complete a user story. They consider the complexity, time, and risk involved rather than the actual hours required. For this project, the Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for estimation because it is widely adopted in Scrum projects.

Story Point Estimation

User Story	Story Points	Justification
Farmer Registration	3	Simple form validation and database storage
Buyer Registration	2	Similar functionality to farmer registration
User Login	3	Secure authentication and session handling
Product Listing	5	Product upload, image handling, and inventory management
Product Search	5	Search and filtering algorithms
Order Placement	8	Multiple validations and database operations
Payment Integration	8	External payment gateway integration
Admin Dashboard	5	User management and reporting features

Justification

Lower story points are assigned to simple tasks such as registration and login, while higher story points are given to features involving complex business logic or external integrations, such as payment processing and order management. Using story points helps the Scrum team estimate workload accurately and plan each sprint effectively.

7. Present the Sprint Goal and Expected Deliverables

7.1 Sprint Goal

The primary goal of Sprint 1 is to develop a functional prototype of the Digital Farmer Marketplace with essential features that allow farmers and buyers to interact through a secure online platform. The sprint focuses on delivering the minimum viable product (MVP), which forms the basis for future development.

Expected Sprint Deliverables

The following deliverables are expected at the end of Sprint 1:

- Farmer Registration Module
- Buyer Registration Module
- Secure Login and Authentication
- Product Listing Module
- Product Search Functionality
- Basic Admin Dashboard
- Database Design and Integration
- Unit Tested Core Features
- User-Friendly Interface

Expected Benefits

- Farmers can register and upload agricultural products.
- Buyers can search and view available products.
- Administrators can manage users and monitor the platform.
- The project team obtains a working prototype for stakeholder review and feedback.
- The completed sprint establishes a strong foundation for implementing advanced features such as online payments, delivery tracking, notifications, analytics, and AI-based recommendations in future sprints.